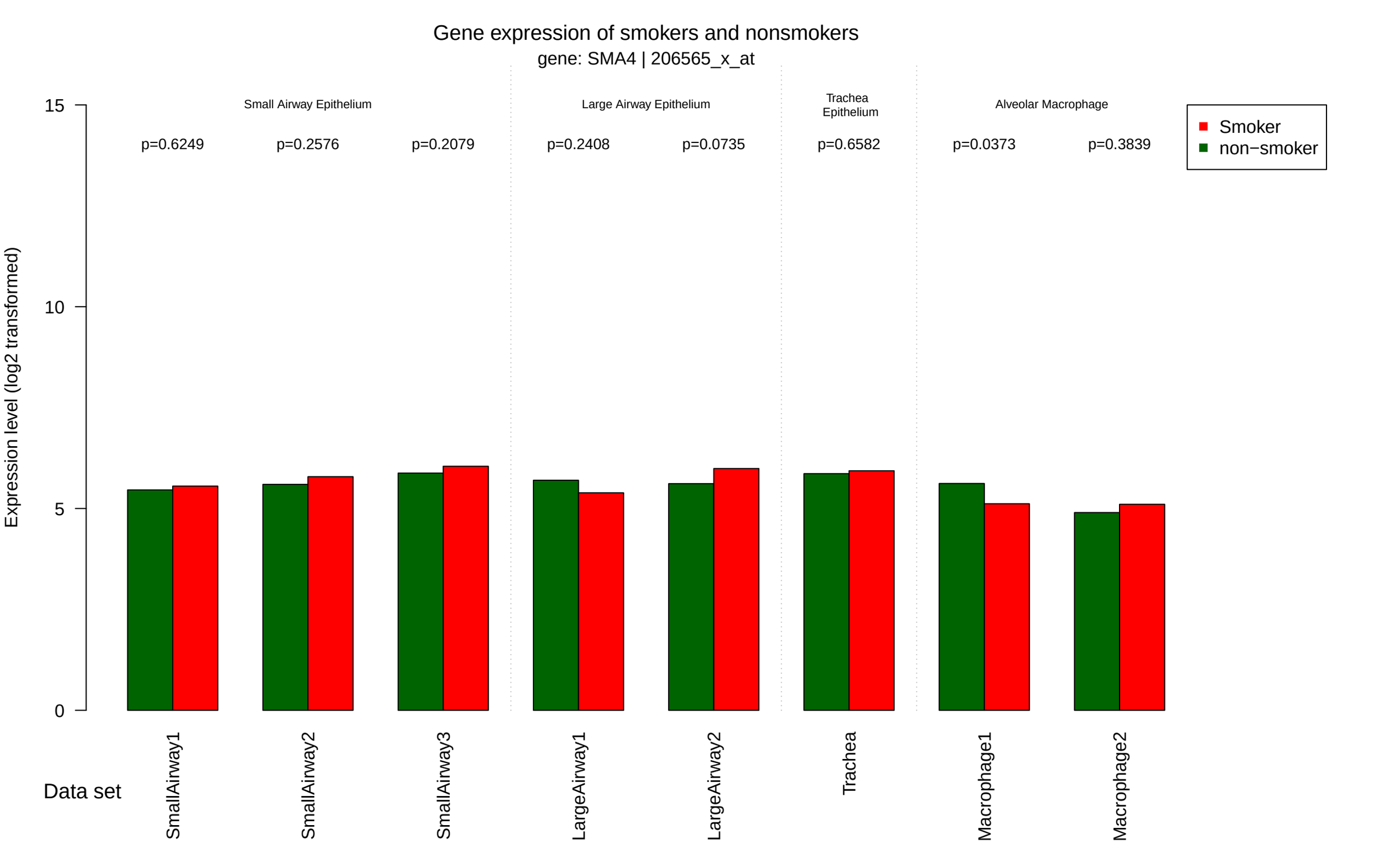
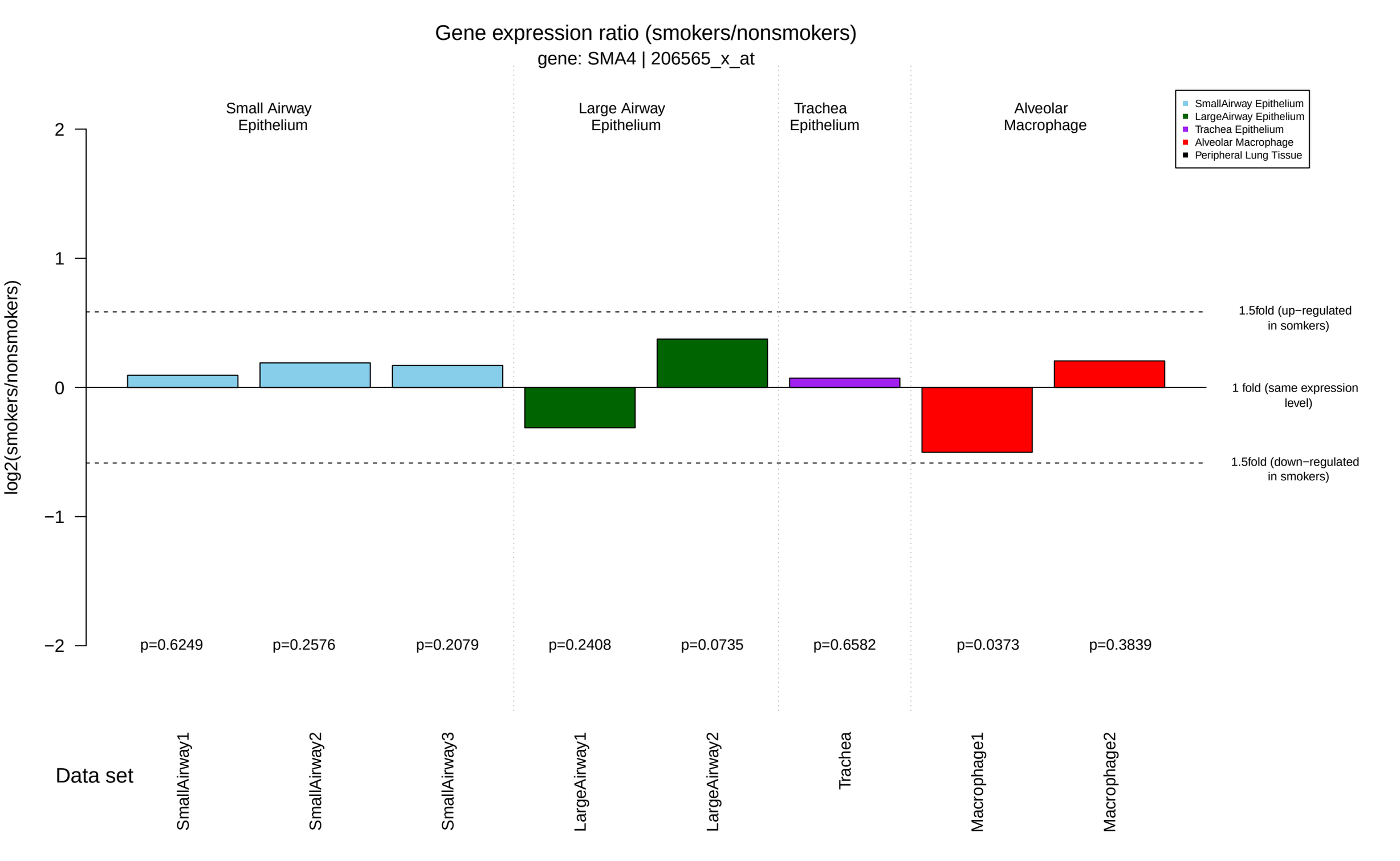
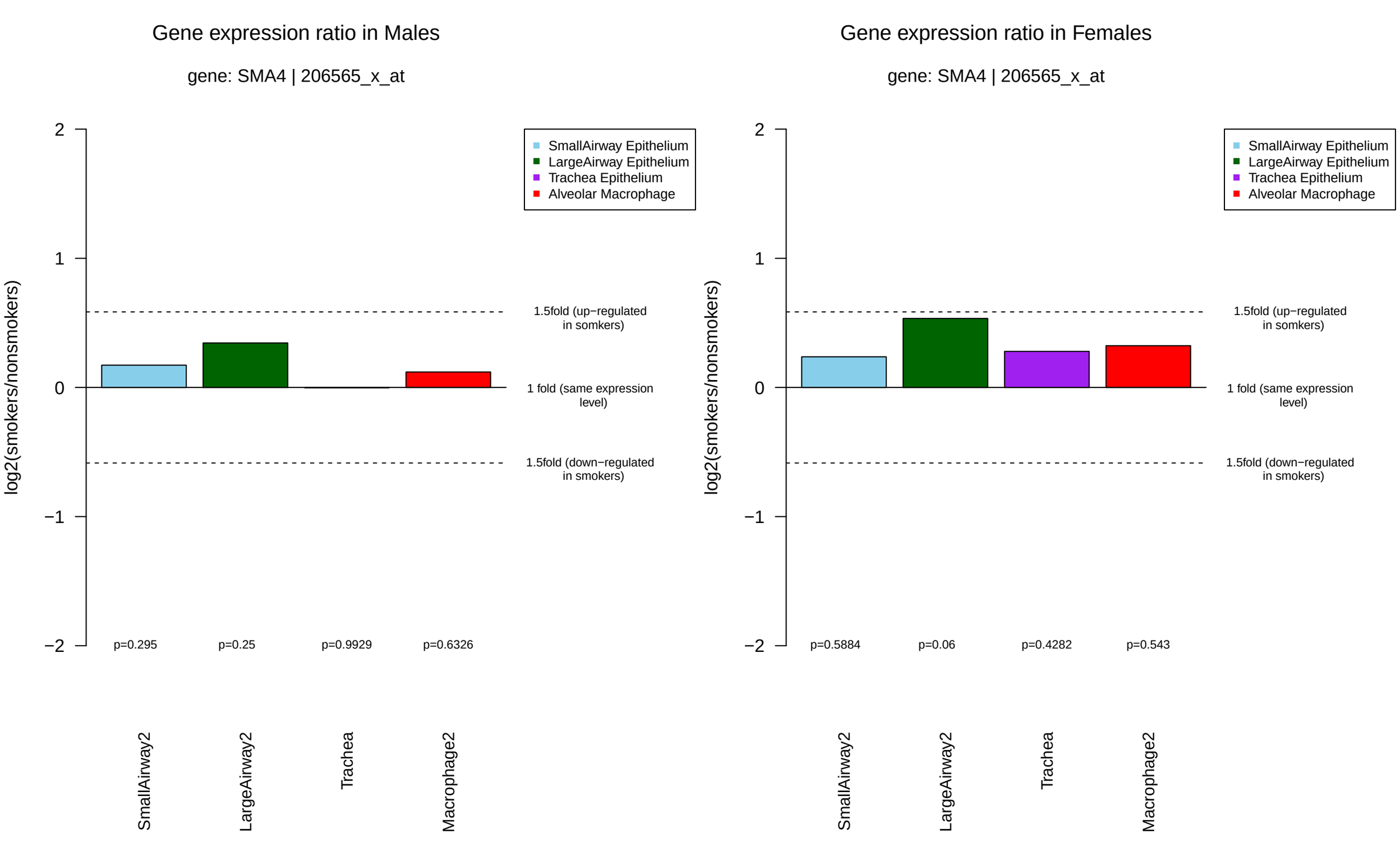
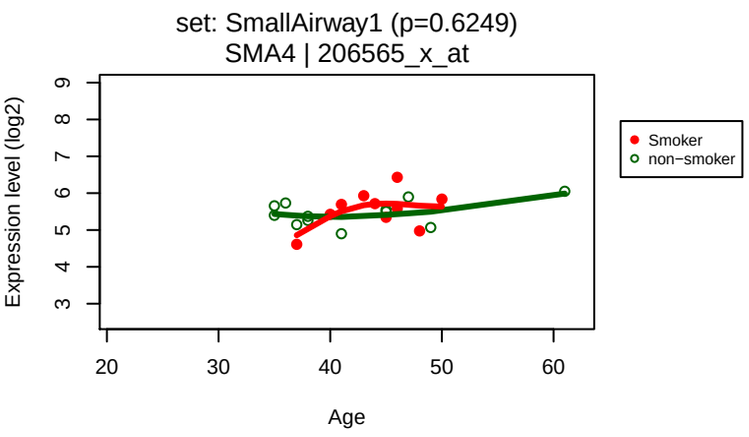
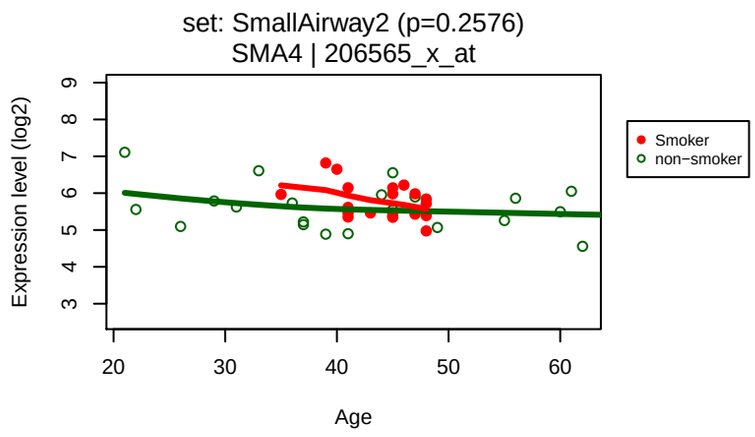
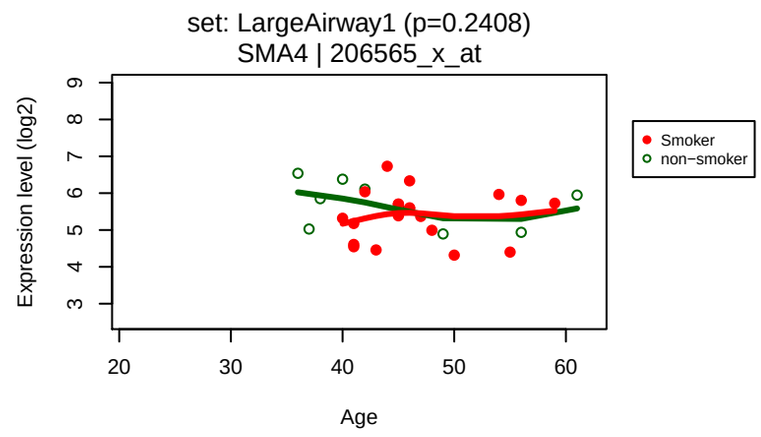

Gene expression of smokers and nonsmokers
gene: SMA4 | 206565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6249
p=0.2576
p=0.2079
p=0.2408
p=0.0735
p=0.6582
p=0.0373
p=0.3839
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

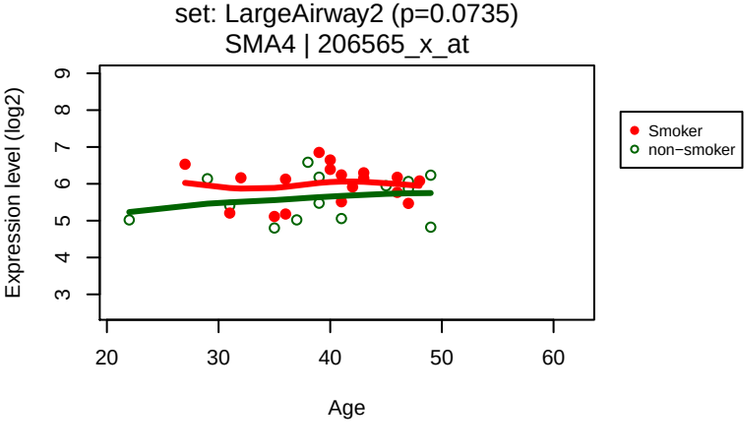

Gene expression ratio (smokers/nonsmokers)
gene: SMA4 | 206565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6249
p=0.2576
p=0.2079
p=0.2408
p=0.0735
p=0.6582
p=0.0373
p=0.3839
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

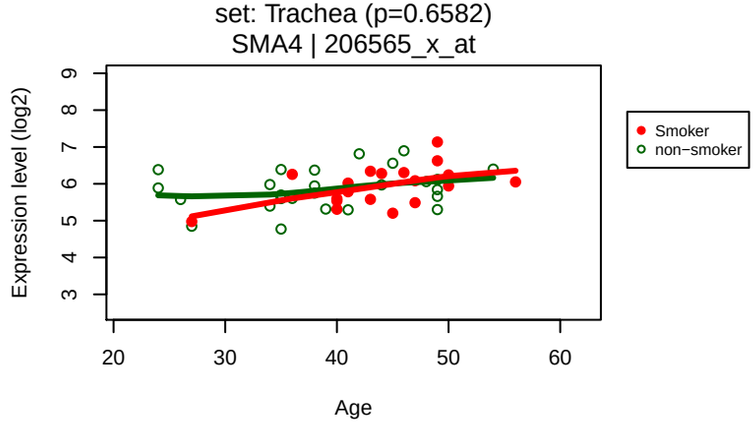

Gene expression ratio in Males
gene: SMA4 | 206565_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.295
p=0.25
p=0.9929
p=0.6326
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMA4 | 206565_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5884
p=0.06
p=0.4282
p=0.543
SmallAirway2
LargeAirway2
Trachea
Macrophage2

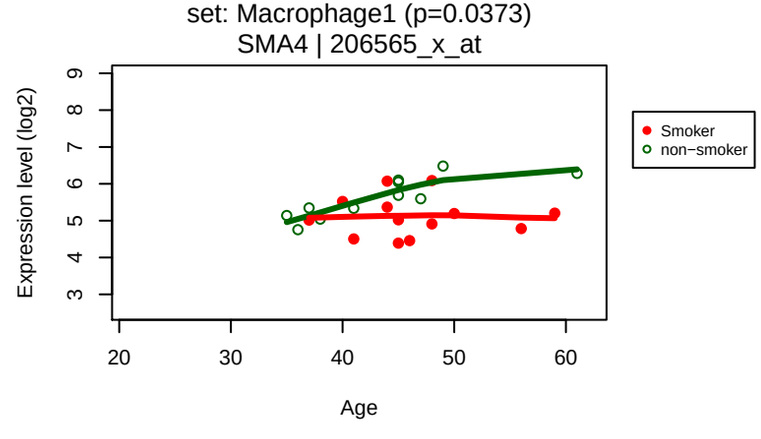

set: SmallAirway1 (p=0.6249)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker

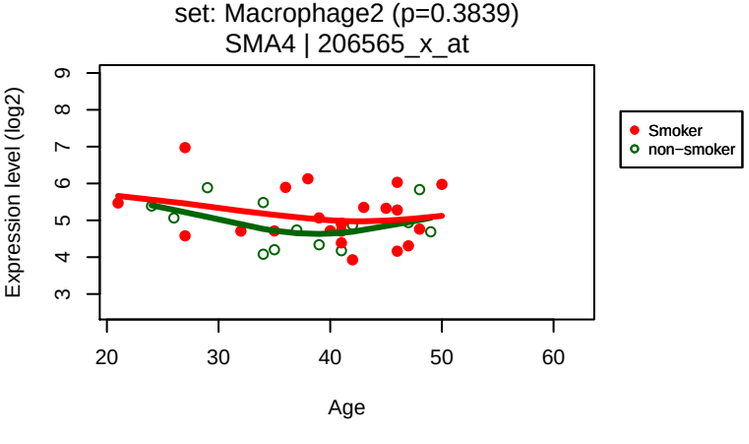

set: SmallAirway2 (p=0.2576)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker

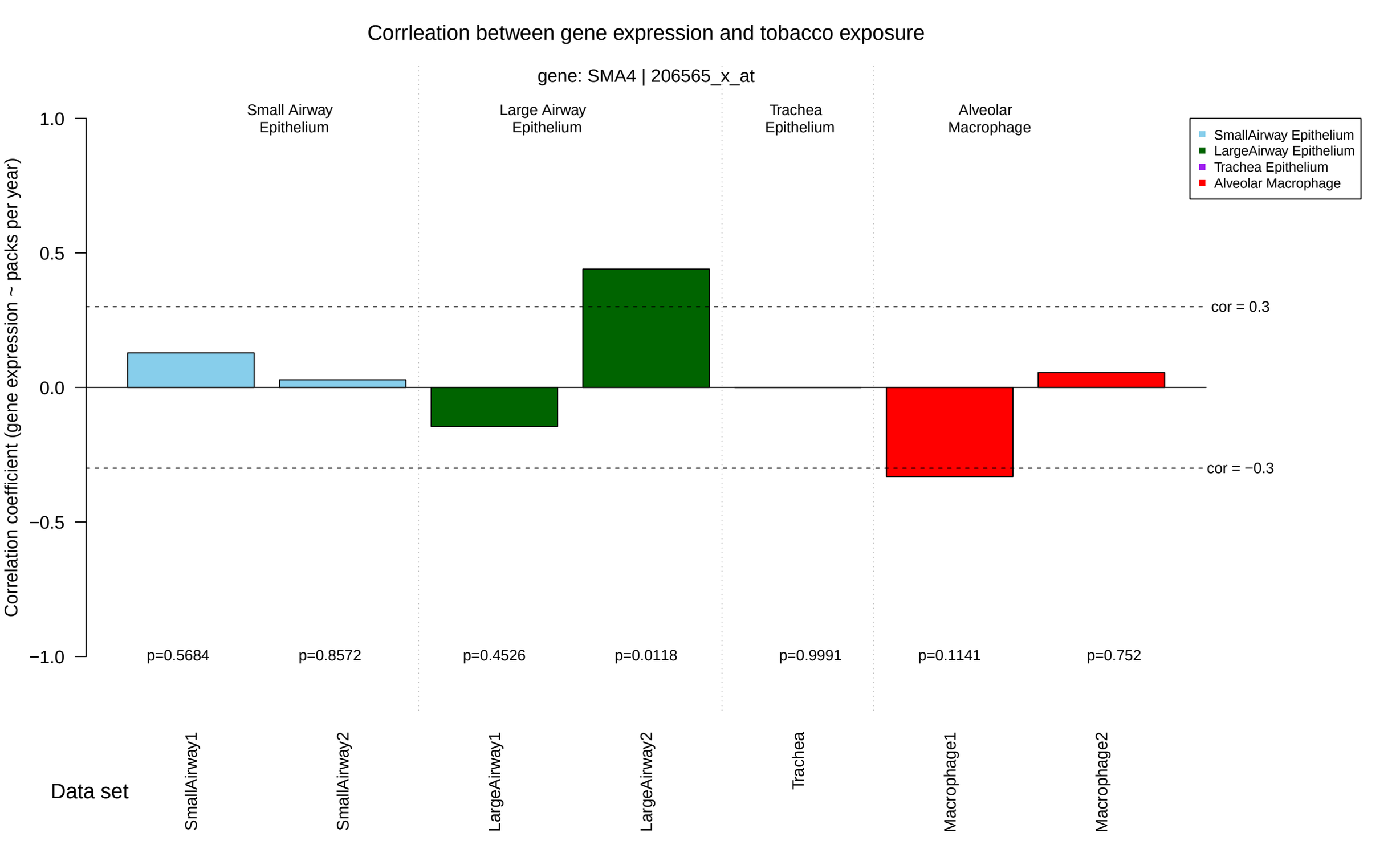

set: LargeAirway1 (p=0.2408)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0735)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6582)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0373)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3839)
SMA4 | 206565_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMA4 | 206565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5684
p=0.8572
p=0.4526
p=0.0118
p=0.9991
p=0.1141
p=0.752
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2